CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

Database Management Systems

Provided by VU Answer

Q1) Syntax of inner and outer join

Answer:

INNER JOIN

SELECT columns

FROM table1

INNER JOIN table2

ON table1.column = table2.column;

OUTER JOIN

SELECT columns

FROM table1

FULL OUTER JOIN table2

ON table1.column = table2.column;

Q2) Syntax of create, select and view commands

Answer:

CREATE TABLE

CREATE TABLE table_name;

SELECT COMMAND

SELECT column_name(s) FROM table_name;

VIEW COMMAND

VUANSWER.PK 0316-2965677 2025

CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

CREATE VIEW view_name AS SELECT column_name(s) FROM table_name;

Q3) Definition of 3rd NF

Answer:

A table is in 3rd Normal Form (3NF) if it is in 2nd Normal Form and all the attributes are functionally dependent only on the primary key. In other words, there are no transitive dependencies.

Q4) What is not true about materialized view, Transactions, and serialized view

Answer:

Materialized View: Not true - It always reflects real-time data.

Transactions: Not true - They cannot be rolled back.

Serialized View: Not true - It allows concurrent execution without any order.

Q5) When DEADLOCK occur

Answer:

Deadlock occurs when two or more transactions are waiting for each other to release locks, and none of them can proceed.

Q6) DCL stand for

Answer:

Data Control Language.

Q7) DML command

Answer:

Data Manipulation Language commands

- INSERT

VUANSWER.PK 0316-2965677 2025

CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

- UPDATE
- DELETE
- SELECT

Q8) Types of Horizontal Partitioning

Answer:

1. **Range Partitioning**: Based on ranges of values.
2. **List Partitioning**: Based on discrete values.
3. **Hash Partitioning**: Based on a hash function.

Q9) Create view of. Employee table where salary greater then 25000

Answer:

```
CREATE VIEW HighSalaryEmployees AS
SELECT *
FROM Employee
WHERE Salary > 25000;
```

Q10) Create view of department table (dept_id,dept-name)

Answer:

```
CREATE VIEW DepartmentView AS
SELECT dept_id, dept_name
FROM Department;
```

Q11) Write SQL command

Answer:

CREATE

CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

CREATE TABLE table_name;

INSERT

INSERT INTO table_name;

UPDATE

UPDATE table_name;

DELETE

DELETE FROM table_name;

Q12) Locking types

Answer:

1. Shared Lock (S)
2. Exclusive Lock (X)
3. Update Lock (U)

Q13) RAID-5 explain

Answer:

RAID-5 uses block-level striping with distributed parity. It offers fault tolerance and improved performance. If one drive fails, the data can be reconstructed using the parity information.

Q14) Write the Types modes of lock.

Answer:

Types of Lock:

- Shared Lock (S)

VUANSWER.PK 0316-2965677 2025

- Exclusive Lock (X)
- Update Lock (U)

Modes of Lock

- **Read Lock**: Allows reading.
- **Write Lock**: Allows writing.

Q15) Write 2 basic types of Disk.

Answer:

1. **HDD (Hard Disk Drive)**
2. **SSD (Solid State Drive)**

Q16) Write the name of partition according to the fallow statement

(1). Which partition is performed on the basis of row.

(2). Which partition is performed on the basis of columns.

Answer:

Horizontal Partition: Based on rows.

Vertical Partition: Based on columns.

Q17) Unique index

Answer:

A unique index ensures that all the values in the index are distinct.

Q18) Which technique maintain the locks

Answer:

Two-phase locking protocol maintains the locks.

VUANSWER.PK 0316-2965677 2025

CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

Q19) Tables define their datatypes

Answer:

```
CREATE TABLE Products (
    Products_ID INTEGER PRIMARY KEY,
    Product_description NUMERIC,
    Product_name TEXT,
    Product_Quality NUMERIC
);
```

Q20) Supposing that the value of X before the addition operation is 23 and after the execution of operation it becomes 33. Now against the write operation, the entry made in the log file will be?

Answer:

If the value of X before the operation is 23 and after the operation is 33, the log entry would be

<X, 23, 33>

Q21) "The normalization process must be done in database" is the given statement is correct or incorrect give valid reason.

Answer:

Yes, normalization is necessary to eliminate data redundancy and ensure data integrity.

Q22) Unique index create

Answer:

```
CREATE UNIQUE INDEX idx_unique_product_name ON Products (Product_name);
```

Q23) Transaction & Locking

CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

Answer:

Transaction management ensures data integrity and consistency. Locking prevents concurrent access issues.

Q24) Interference Rule

Answer:

Interference rules ensure that concurrent transactions do not cause data inconsistencies.

Q25) DIF F ERENTIATE B/W ALTERNATE AND S ECONDARY KEY WITH EXAMPLES

Answer:

Alternate Key: An alternate key is any candidate key that is not chosen as the primary key. For example, in a table Student with StudentID, Email, and PhoneNumber as unique attributes, if StudentID is the primary key, Email and PhoneNumber are alternate keys.

Secondary Key: A secondary key is a non-primary key that is used for data retrieval. For example, in the Student table, if StudentID is the primary key, and queries often use Email for lookups, Email can be considered a secondary key.

Q26) S QL INS ERT COMMANDS

Answer:

INSERT INTO table_name (column1, column2, column3, ...)

VALUES (value1, value2, value3, ...);

Q27) RAID LEVEL 1, IF EITHER DRIVE F AILS , NO DATA IS LO□T.

THE GIVEN S TATEMENT IS CORRECT OR INCORRECT

Answer:

The given statement is correct.

CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

RAID 1 uses mirroring, meaning data is duplicated on two or more drives. If one drive fails, the data is still accessible from the other drive(s).

Q28) WRITE TWO PRIMARY MODES F OR TAKING LOCKS . AND DIF F ERENTIATE THEM.

Answer:

- Shared Lock (S):
- Exclusive Lock (X)

Difference

Shared locks allow concurrent read access, whereas exclusive locks prevent all other access.

Q29) Difference Candidate key and primary key?

Answer:

Candidate Key: Any key that can uniquely identify a record in a table. There can be multiple candidate keys.

Primary Key: The chosen candidate key to uniquely identify records in a table. There is only one primary key.

Q30) Importance of hash attributed?

Answer:

Hash attributes are important for fast data retrieval and distribution in hash-based indexing and partitioning.

They ensure that data is evenly distributed across partitions and provide constant-time complexity for lookup operations.

Q31) Correct incorrect statements chack points and deadlock?

Answer:

CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

Deadlock occurs when two or more transactions wait indefinitely for each other to release locks.

Correct statements about transactions and locking mechanisms can prevent deadlock.

Q32) SQL query create table

Answer:

```
CREATE TABLE Products (
    ProductID INT PRIMARY KEY,
    ProductDescription VARCHAR(255),
    ProductName VARCHAR(100),
    ProductQuality INT
);
```

Q33) Explain domain and explain attributed?

Answer:

Domain: A domain is a set of valid values that an attribute can take. For example, the domain of an attribute ‘Age’ could be all integers from 0 to 120.

Attribute: An attribute is a column in a table, representing a specific data point. For example, ‘Name’ in a ‘Person’ table.

Q34) Cluster indexes

Answer:

Cluster Indexes these indexes sort the data rows in the table based on the key values.

Q35) Denormalization and normalization

Answer:

Normalization: Process of organizing data to reduce redundancy.

CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

Denormalization: Combining normalized tables to improve query performance by reducing joins.

Q36) Views ka topic main difference b/w dynamic view and complex view

Answer:

Dynamic View: A view that reflects the real-time data from the base tables. It is not stored physically.

Complex View: A view that involves multiple tables, aggregations, or functions. It may or may not be updatable.

Q37) DFD bnana tha 'ordering system ' of McDonald js ma customer order deta or usy bill recipient Di jati JB bil pay krta tha order kitchen ma beja jata. Or bil ki copy supervisor K pass b jati.

Answer:

1. **Customer**: Places Order -> Ordering System
2. **Ordering System**: Sends Bill -> Customer
3. **Customer**: Pays Bill -> Ordering System
4. **Ordering System**: Sends Order Details -> Kitchen
5. **Ordering System**: Sends Bill Copy -> Supervisor

Q38) Define Replication.

Answer:

Replication is the process of copying and maintaining database objects, such as tables, in multiple database instances for redundancy, fault tolerance, and load balancing.

Q39) Differentiate between complex view and materialized view.

Answer:

Complex View: Does not store data, retrieves it dynamically. Example: joining multiple tables.

CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

Materialized View: Stores the result of a query physically. It can be refreshed periodically.

Q40) Write the advantages and disadvantage of Re-Hash and chaining.

Answer:

Re-Hashing

- **Advantages**: Spreads data more evenly, reduces collisions.
- **Disadvantages**: More complex, higher computational cost.

Chaining

- **Advantages**: Simple to implement, easy to handle collisions.
- **Disadvantages**: Can lead to long chains, increased search time.

Q41) Create, drop or view command

Answer:

CREATE

CREATE TABLE table_name;

DROP

DROP TABLE table_name;

VIEW

CREATE VIEW view_name AS SELECT * FROM table_name;

VUANSWER.PK 0316-2965677 2025

CS403 CURRENT SOLVED SUBJECTIVE FINALTERM PAPERS

Note:

For more subjects latest and currently updated files or need help/guidance contact us

Get Most Important PAST PAPERS freely from VUAnswer.pk

Current paper is most important and recommended there will be highly chance to ask in exams as repeated questions.

VUANSWER.PK
0316-2965677
2025

REGARD: SIR SARIM

WHATSAPP +923162965677

Share with fellows and help others in their studies.